Affine Transformations: Hidden Exercise in Linear Algebra

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※ Introduction

Affine transformations are a beautiful topic consisting of many illustrious examples when taken in the context of competitive mathematics. A simple affine transformation has the ability to transform a beast to rather a child. In this handout, we will be going over what an affine transformation is, and then we will apply this technique to various problems.

※ What is an Affine Transformation?

Affine Transformations are any transformation that preserve collinearity. Examples of these include translations, scalings, homothety, similarity, reflections, rotations, hyperbolic rotations, shear mappings, and compositions of them.

It is important to be able to recall these as noticing an instance of an affine transformation allows us to know various properties that are preserved. With this comes the following effects:

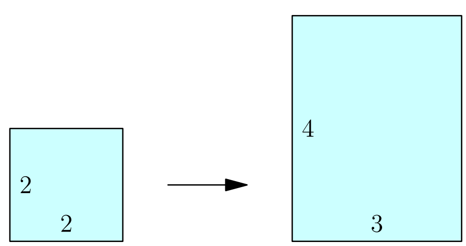
- Proportions of distances are preserved. That is, under an affine transformation $t : (A, B, C) \to (A', B', C')$, then the following ratio is satisfied:
$$\frac{AB}{A'B'} = \frac{BC}{B'C'} = \frac{CA}{C'A'},$$
where A, B, and C are collinear.
- Parallel lines are preserved, along with perpendicular lines.
- Angles are not necessarily preserved.

2.1 Definition

An affine transformation $t : \mathbb{R}^n \to \mathbb{R}^n$ can be expressed as
$$y = t(x) = Ax + B$$
for a $n \times n$ matrix A and a vector $B \in \mathbb{R}^n$.

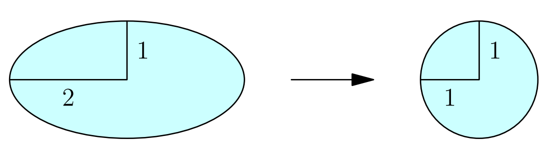
2.2 Instances of Affine Transformation

Square to Rectangle



In particular, we can describe such a transformation by
$$t(x) = \begin{bmatrix} \frac{3}{2} & 0 \\ 0 & 2 \end{bmatrix} x.$$

Ellipse to Circle



In particular, we can describe such a transformation by

$$t(x) = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} x.$$

2.3 Proofs of Properties

Proposition. Under an affine transformation, collinearity is preserved.

Proof. Let $A \in \mathbb{R}^n$ and $B \in \mathbb{R}^n$. Define $C = A + c(B - A)$ for some constant c. Then, for some transformation $t(x) = Mx + N$, it follows that

$$\begin{aligned} t(C) = t(A + c(B - A)) &= (A + c(B - A)) \cdot M + N \\ &= (A \cdot M + N) + c \cdot (B \cdot M - A \cdot M) \\ &= (A \cdot M + N) + c \cdot \left[(B \cdot M + N) - (A \cdot M + N)\right] \\ &= t(A) + c \cdot \left[t(B) - t(A)\right], \end{aligned}$$

and so the transformed values preserve collinearity, as desired. ■

Proposition. Under an affine transformation, proportions of distances are preserved.

Proof. We will show the following: under an affine transformation $t : (A, B, C) \longrightarrow (A', B', C')$, then the following ratio is satisfied:

$$\frac{AB}{A'B'} = \frac{BC}{B'C'} = \frac{CA}{C'A'},$$

where A, B, and C are collinear. Let $t(x) = Mx + N$ for a $n \times n$ matrix M and $N \in \mathbb{R}^n$. Then,

$$A'B' = B' - A' = (MB + N) - (MA + N) = MB - MA = M(B - A) = MAB.$$

Likewise, $C' - A' = M(C - A)$. Suppose $k = \frac{AB}{AC}$. Then, by definition, then $B - A = k \cdot (C - A)$. Substituting, then it follows that

$$\begin{aligned} B - A = k \cdot (C - A) &\iff (B' - A') \cdot M^{-1} = k \cdot \left[(C' - A') \cdot M^{-1}\right] \\ &\iff B' - A' = k \cdot (C' - A') \\ &\iff A'B' = k = A'C'. \end{aligned}$$

Hence, $\frac{AB}{AC} = \frac{A'B'}{A'C'}$, as desired. ■

Proposition. Under an affine transformation, parallel lines are preserved.

Proof. Consider the transformation $t(x) = Mx + N$. Consider lines

$$\begin{cases} \ell_1 : y(x) = xv + c_1, \text{ and} \\ \ell_2 : y(x) = xv + c_2. \end{cases}$$

for a vector $v \in \mathbb{R}^n$. Then,

$$t(\ell_1) = M(xv + c_1) + N = x \cdot Mv + (Mc_1 + N), \quad \text{and}$$
$$t(\ell_2) = M(xv + c_2) + N = x \cdot Mv + (Mc_2 + N).$$

Both transformed lines $t(\ell_1)$ and $t(\ell_2)$ now share the same direction vector Mv, as desired. ■

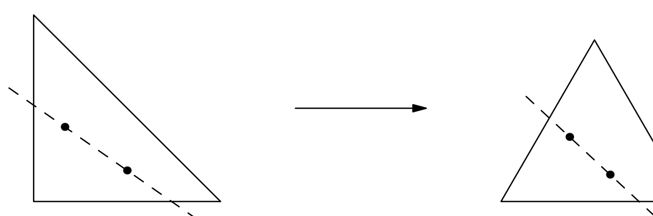
※ Walkthroughs

3.1 Instagram

(**Instagram**) Two points P and Q are selected in square $ABCD$ at random. What is the probability that the line PQ intersects AB and BC?

Proof. Evidently, the points P and Q must lie inside triangle ABC. This occurs with probability $\frac{1}{4}$.

Following this, triangle ABC is an isosceles right triangle, we can take an affine transformation to transform it to an equilateral triangle:



For an equilateral triangle, the probability that the two points form a line intersecting $A'B'$ and $B'C'$ is $\frac{1}{3}$ by symmetry. Multiplying together yields

$$\frac{1}{4} \cdot \frac{1}{3} = \boxed{\frac{1}{12}.}$$

3.2 2020 PUMaC Geometry, Daniel Carter

(**2020 PUMaC**, Daniel Carter) Triangle $\mathcal{T}$ with side lengths 13, 14, and 15 is inscribed in an ellipse $\mathcal{E}$. Compute the maximum possible area of $\mathcal{E}$.

Proof. Take an equilateral triangle $\triangle$ with circumradius 1. Note, the area of the triangle is $\frac{3\sqrt{3}}{4}$ and the area of the circumcircle Ω is π.

Under an affine transformation from the equilateral triangle to $\mathcal{T}$, we know that proportions on areas are conserved. Hence, as the area of $\mathcal{T}$ is 84, it follows that

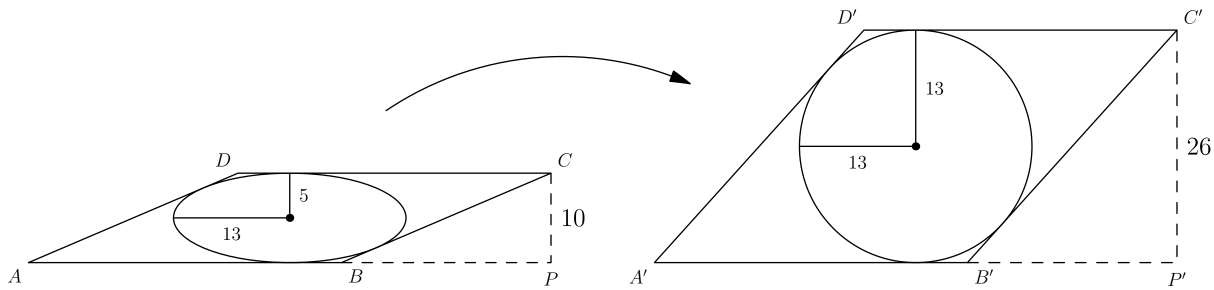
$$\frac{[\Omega]}{[\mathcal{E}]} = \frac{[\triangle]}{[\mathcal{T}]} \implies \frac{\pi}{\mathcal{E}} = \frac{3\sqrt{3}/4}{84} = \frac{3}{112\sqrt{3}} \implies [\mathcal{E}] = \boxed{\frac{112\sqrt{3}\pi}{3}}.$$

3.3 2023 ZIME, Oron Wang

(**2023 ZIME**, Oron Wang) Parallelogram $ABCD$ has an area of 350 and satisfies $AB = 35$. Let F and G be points in the interior of the parallelogram such that $FG = 24$ and $FG \parallel AB$. If there exists an ellipse with foci F and G tangent to all four sides of the parallelogram, find BC.

Proof. Note the height of the parallelogram is 10. As the minor axis is 10 and distance between the two foci is 24, then the major axis has length of 26.

Take an affine transformation to transform an ellipse to a circle. In general, this is usually a good idea because a circle is a far more comfortable object than an ellipse.



The transformation taken in this instance is

$$y = t(x) = \begin{bmatrix} \frac{13}{5} & 0 \\ 0 & 1 \end{bmatrix} x.$$

Under the transformation, then $P'C' = PC \cdot \frac{13}{5} = 26$. In addition, as $ABCD$ is a parallelogram and parallelism is conserved, then $A'B'C'D'$ is a parallelogram, as well. However, a parallelogram with an inscribed circle is a rhombus. Hence, $B'C' = A'B' = AB = 35$. Hence,

$$BP = B'P' = \sqrt{B'C'^2 - P'C'^2} = \sqrt{35^2 - 26^2} = \sqrt{549}.$$

So, from Pythagorean Theorem, then $BC = \sqrt{BP^2 + CP^2} = \sqrt{549 + 10^2} = \boxed{\sqrt{649}.}$

※ Problems

1. (**Ze Math Competitions**, Benny Wang) Celine draws a regular hexagon $ABCDEF$ in the Cartesian plane such that $\overline{AB}$ and $\overline{DE}$ are parallel to the x-axis. She then stretches the plane vertically by a factor of some constant k and draws the image of the hexagon under this stretch, $A'B'C'D'E'F'$. If the distance between lines $A'B'$ and $D'E'$ is 5 units and the distance between lines $B'C'$ and $E'F'$ is 2 units, what is k?

2. (**Art of Problem Solving**) Let ABC be a triangle with orthic triangle DEF. Compute, with proof, the value of
$$\frac{[ABC]}{[DEF]}.$$

3. (**Art of Problem Solving**) Triangle ABC has interior point D. Lines AD, BD, CD intersect BC, AC, AB, respectively, at X, Y, Z, respectively. Suppose that A', B', C' are the reflections of D across points X, Y, Z, respectively. Prove that points A, B, C, A', B', C' lies on an ellipse.

4. (**2025 HMMT**, Albert Wang) Two points are selected independently and uniformly at random inside a regular hexagon. Compute the probability that a line passing through both of the points intersects a pair of opposite edges of the hexagon.

5. (**2020 HMMT**) Let $ABCD$ be a parallelogram. Points X and Y lie on segments AB and AD respectively, and AC intersects XY at point Z. Prove that
$$\frac{AB}{AX} + \frac{AD}{AY} = \frac{AC}{AZ}.$$

6. (**2019 ISL**) Let P be a point inside triangle ABC. Let AP meet BC at A_1, let BP meet CA at B_1, and let CP meet AB at C_1. Let A_2 be the point such that A_1 is the midpoint of PA_2, let B_2 be the point such that B_1 is the midpoint of PB_2, and let C_2 be the point such that C_1 is the midpoint of PC_2. Prove that points A_2, B_2, and C_2 cannot all lie strictly inside the circumcircle of triangle ABC.

7. (**2015 Bulgaria**) In a triangle $\triangle ABC$ points L, P and Q lie on the segments AB, AC and BC, respectively, and are such that $PCQL$ is a parallelogram. The circle with center the midpoint M of the segment AB and radius CM and the circle of diameter CL intersect for the second time at the point T. Prove that the lines AQ, BP and LT intersect in a point.

References

[1] Weisstein, Eric W. "Affine Transformation.". MathWorld, Wolfram. Online; accessed May 22, 2026.
https://mathworld.wolfram.com/AffineTransformation.html

[2] "Affine Transformation". ORMC, UCLA Math Circle. Online; accessed May 22, 2026.
https://circles.math.ucla.edu/circles/lib/data/Handout-4445-4280.pdf